Adaptations of the Small Intestine

Exam code: 0610 & 0970

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Updated on 3 October 2024

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Adaptations of the Small Intestine: Extended

Extended Tier Only

- The ileum is adapted for absorption as it is **very long** and has a **highly folded surface with millions of villi** (tiny, finger like projections)
- These adaptations massively **increase the surface area** of the ileum, allowing absorption to take place faster and more efficiently

Help me understand

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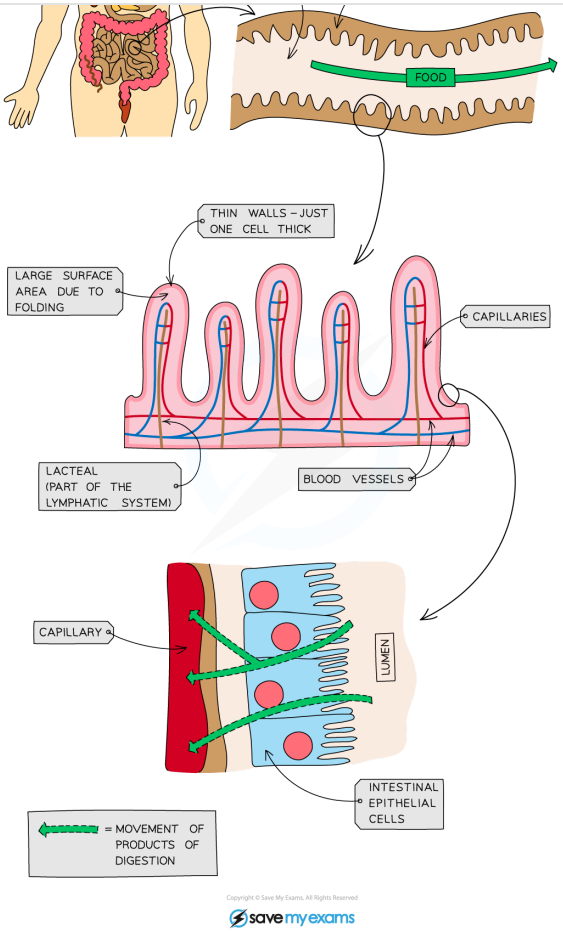

Adaptations of the small intestine

- **Microvilli** on the surface of the villus further increase surface area for faster absorption of nutrients
- Wall of the villus is **one cell thick** meaning that there is only a short distance for absorption to happen by diffusion and active transport
- Well supplied with a **network of blood capillaries** that transport glucose and amino acids away from the small intestine in the blood
- **Lacteal** runs through the centre of the villus to transport fatty acids and glycerol away from the small intestine in the lymph

Examiner Tips and Tricks

The **way in which the structure of a villus is related to its function** comes up frequently in exam questions so it is worth ensuring you have learned these adaptations.

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